(Please write your Exam Roll no.) Exam Roll No. .....

MID TERM EXAMINATION

SIXTH SEMESTER [B.TECH.] APRIL 2022

Paper Code: ETEC -306 **Subject: Digital Signal Processing**

Time: 1Hour 30 Min. **Maximum Marks: 30**

Note: Attempt any three questions including Q. No. 1 which is compulsory.

Q1. (a) Compute the 4-point DFT of the sequence, $x(n) = \cos(n\frac{\pi}{2})$. [2]

(b) Illustrate the properties of Twiddle Factor. [2]

(c) Why Bit-reversal concept is important in FFT algorithms? [2]

(d) What are the conceptual differences between FIR and IIR filters. [2]

(e) List the advantages and disadvantages of digital filters. [2]

Q2. (a) Let $x(n) = \{1, 1, 1, 1\}$ and $h(n) = \{1, 1, 1, 1\}$. Evaluate circular convolutions for (i) N=7, (ii) N=4. [5]

(b) Compute the 4-point FFT of the sequence $x(n) = \{0,1,2,3\}$ using DIT-FFT algorithm. Draw the signal flow graph. [5]

Q3. Design an IIR low pass Butterworth filter using bilinear transformation to meet the following specifications.

Passband: $0.8 \leq |H(e^{j\omega})| \leq 1$ for $0 \leq \omega \leq 0.2\pi$

Stopband: $|H(e^{j\omega})| \leq 0.2$ for $0.6\pi \leq \omega \leq \pi$ [10]

Q4. (a) Realize the given transfer function in cascade form. [5]

$$H(z) = \frac{5z\,(z+0.4)}{(z-0.2)(z-0.6)}$$

(b) Let $x(n) = \{n+1\}, 0 \leq n \leq 9$ and $h(n) = \{1, 0, -1\}$. Implement the overlap-save method to compute $y(n) = x(n) \otimes h(n)$. [5]

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